

# Titouan Guerin

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## Education

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**Sorbonne Université**, MSc in AI & Machine Learning Sept 2023 – Present

- Focus on the mathematical foundations of Machine Learning
- **Classes:** Statistics & Probabilities, Advanced ML, Deep Learning, GenAI, Information Retrieval...

**University of Luxembourg**, BS in Computer Science Sept 2020 – Aug 2023

- Member of the Competitive Programming Club (participated in BAPC 2021)
- **Classes:** Linear Algebra, Algorithms & Complexity, Advanced Programming, Data Science ...

## Experience

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**AI Research Assistant**, Uni Lu – Luxembourg July 2024 – Aug 2024

- Developed the backend of a multi-agent system learning app powered by LLMs using LangGraph, designed for low-resource language learning (Luxembourgish)
- Published work at *PRIMA conference in Kyoto, Japan*

**Data Analysis Intern**, Grant Thornton – Luxembourg Aug 2022 – Sep 2022

- Interned in the Security Department at GT LU, where I performed data analysis on the internal servers
- Developed VBA Add-on for the outlook service to optimize priority email filtering for employees and clients

## Publications

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**Beyond Chatbots: Enhancing Luxembourgish Language Learning Through Multi-agent Systems and Large Language Model** Nov 2024

Sana Nouzri, Meryem EL Fatimi, *Titouan Guerin*, Mahfoud Othmane, Amro Najjar

PRIMA 2024: Principles and Practice of Multi-Agent Systems, [Read here](#)

## Projects

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**Reinforcement Learning for Dark Souls** – [github.com/Titouaaaaan/RL-Souls](https://github.com/Titouaaaaan/RL-Souls)

- Implemented from scratch RL algorithms to train an agent to beat bosses in the video game *Dark Souls III*
- Tools Used: Gymnasium, Pytorch, Tensorboard, Python

**From Scratch Neural Networks** – [github.com/Titouaaaaan/DIY-Neural-Network](https://github.com/Titouaaaaan/DIY-Neural-Network)

- Implemented Neural Networks from scratch (using only numpy) in a modular way.
- Features: Activation Functions (TanH, Sigmoid, Softmax...), Linear/Pooling/Convolution Layers, Loss Functions (MSE, Cross-Entropy, Categorical CE...), SGD Optimizer, PCA/T-SNE, K-Means
- Tools: Python, Numpy, Pandas

**AI Chess Coaching** – [github.com/Titouaaaaan/ChessCoach](https://github.com/Titouaaaaan/ChessCoach)

- Full-stack app that integrates Stockfish engine for gameplay and uses an AI-powered assistant to analyze, comment on games in real-time and guide the player to get better
- Tools: Python, Javascript, CSS, FastAPI, NPM, LangGraph

## Technologies

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**Languages:** Python, Javascript, Java, SQL, CSS

**Technologies:** PyTorch, TensorFlow, Gymnasium, LangGraph, React, NPM, Git, FastAPI